

From: Robbins and Cotran, Pathologic Basis of Disease, 11th edition, 2025
METASTATIC TUMORS TO THE BRAIN

Metastatic lesions, mostly carcinomas, account for approximately one-quarter to one-half of intracranial tumors in hospitalized patients. The five most common primary sites are lung, breast, skin (melanoma), kidney, and gastrointestinal tract, accounting ...The meninges are also a frequent site of involvement by metastatic disease, where they may either form discrete masses or a more disseminated subarachnoid, meningitis-like pattern known as *meningeal carcinomatosis* ;... Metastatic tumors present clinically as mass lesions and may occasionally be the first manifestation of systemic cancer. In general, localized treatment of solitary brain metastases improves the quality and duration of the patient's life.

MORPHOLOGY

Intraparenchymal metastases form sharply demarcated masses, often at the junction of gray and white matter, usually surrounded by a zone of edema. The boundary between tumor and brain parenchyma is usually well-defined microscopically...The tumors typically resemble their parent tumor type but can be very poorly differentiated in some cases; for the roughly 10% that present as a metastasis of unknown primary, immunohistochemistry is often helpful for further subtyping. Meningeal carcinomatosis, with tumor nodules studding the surface of the brain, spinal cord, and intradural nerve roots, is most often associated with carcinoma of the lung and breast.

Familial tumor syndromes

Neurofibromatoses

The two forms of *neurofibromatosis* , NF1 and NF2, are biologically distinct familial **autosomal dominant syndromes** characterized by tumors of the peripheral and central nervous systems. NF1 has a frequency of 1 in 3000 individuals and is characterized by neurofibromas of peripheral nerves, gliomas of the optic nerve, pigmented nodules of the iris (Lisch nodules), and cutaneous hyperpigmented macules (café au lait spots).

NF2 is less common (with a frequency of 1 in 40,000 to 50,000 persons) and is characterized by the occurrence of bilateral vestibular (cranial nerve VIII) schwannomas, multiple meningiomas, and ependymomas of the cervical spinal cord.

Tuberous sclerosis complex

Tuberous sclerosis complex (TSC) is an autosomal dominant syndrome that occurs at a frequency of approximately 1 in 6000 births. It is characterized by the development of hamartomas and benign neoplasms involving the brain and other tissues; the most frequent clinical manifestations are seizures, autism, and intellectual disability. Hamartomas within the CNS take the form of cortical tubers and subependymal nodules; **subependymal giant cell astrocytomas (SEGA)** are benign neoplasms that appear to develop from the hamartomatous nodules in the same locations. Cortical tubers are often epileptogenic, and surgical resection can be beneficial when medical management of the seizures fails.

Elsewhere in the body, **renal angiomyolipomas**, retinal glial hamartomas, pulmonary lymphangiomyomatosis, and **cardiac rhabdomyomas** develop during childhood and

adolescence. Cysts may be found at various sites, including the liver, kidneys, and pancreas. Cutaneous lesions include **angiofibromas, localized leathery thickenings (shagreen patches), hypopigmented areas (ash-leaf patches), and subungual fibromas.**

One tuberous sclerosis gene (*TSC1*) is found on chromosome 9q34 and encodes a protein known as hamartin; the more commonly mutated gene (*TSC2*) is on 16p13.3 and encodes tuberin. These two proteins form a complex that inhibits the kinase mTOR, a key regulator of protein synthesis and other aspects of anabolic metabolism. Mutations in *TSC1* or *TSC2* disrupt this control and lead to increased and unregulated mTOR activity... Treatment is usually symptomatic, including anticonvulsant therapy for control of seizures; however, treatment with mTOR inhibitors has sometimes resulted in dramatic clinical improvement.

MORPHOLOGY

Cortical hamartomas of TSC are firm areas that, in contrast to the softer adjacent cortex, have been likened to potatoes (hence the appellation “**tubers**”). The tubers are composed of haphazardly arranged neurons that lack the normal laminar organization. In addition, some dysplastic cells have an appearance intermediate between glia and neurons (large vesicular nuclei with nucleoli, resembling neurons, and abundant eosinophilic cytoplasm resembling gemistocytic astrocytes); they often express intermediate filaments of both neuronal (neurofilament) and glial (GFAP) types. Similar hamartomatous features are present in the subependymal nodules, where the large astrocyte-like cells cluster beneath the ventricular surface. These *small* wax-like masses that bulge into the ventricular system gave rise to the term **candle gutterings**. Subependymal giant cell astrocytoma is histologically similar, but larger; it often presents with obstructive hydrocephalus.

Von hippel-lindau disease

Individuals with this autosomal dominant disease develop hemangioblastomas of the CNS; cysts that involve the pancreas, liver, and kidneys; renal cell carcinomas; and pheochromocytomas. Hemangioblastomas are most common in the cerebellum and retina but may also occur in other CNS locations. The disease frequency is 1 in 30,000 to 40,000 individuals. The gene associated with von Hippel-Lindau disease, called *VHL*, is a tumor-suppressor gene located on chromosome 3p25.3; it encodes a protein (VHL) that is a component of a ubiquitin ligase complex that down-regulates transcription factors of the hypoxia-induced factor (HIF) family. HIFs regulate the expression of VEGF, erythropoietin, and other growth factors. VEGF leads to excessive vessel growth, contributing to the development of hemangioblastomas. Erythropoietin overexpression leads to polycythemia in about 10% of hemangioblastoma patients. ...Therapy is directed at the symptomatic neoplasms, including resection of the cerebellar hemangioblastomas and laser therapy for retinal hemangioblastomas.

MORPHOLOGY

Hemangioblastomas are highly vascular neoplasms that form a mural nodule associated with a large fluid-filled cyst. The lesion consists of numerous capillary-size or somewhat larger thin-walled vessels with intervening neoplastic cells that have vacuolated, lipid-rich cytoplasm. The neoplastic stromal cells express erythropoietin, VEGF, and inhibin (a diagnostically useful marker), and are believed to be derived from an early mesenchymal progenitor cell capable of differentiating into both endothelial and hemopoietic cells.